

switch to it with `git checkout contact-form`. This keeps your work separate from the `main` branch until it's ready.

Merge: Merging integrates changes from one branch into another. Git automatically handles many merges, but conflicts may need manual resolution.

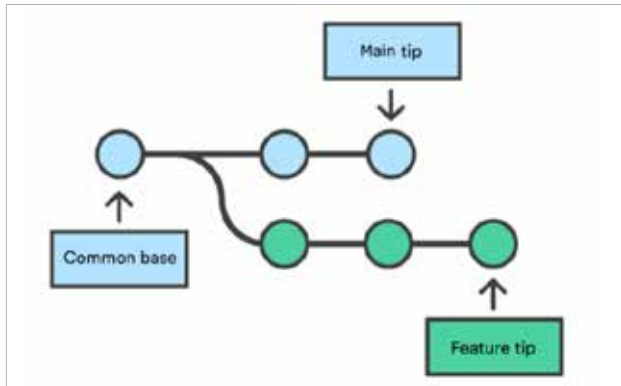


Figure 3: Merging branch

Once the `contact-form` feature is complete and tested, you can merge it into the `main` branch using `git checkout main` followed by `git merge contact-form`. If there are no conflicts, Git combines the changes automatically.

Pull request (PR): A pull request is a request to merge changes from one branch into another, usually accompanied by a review process.

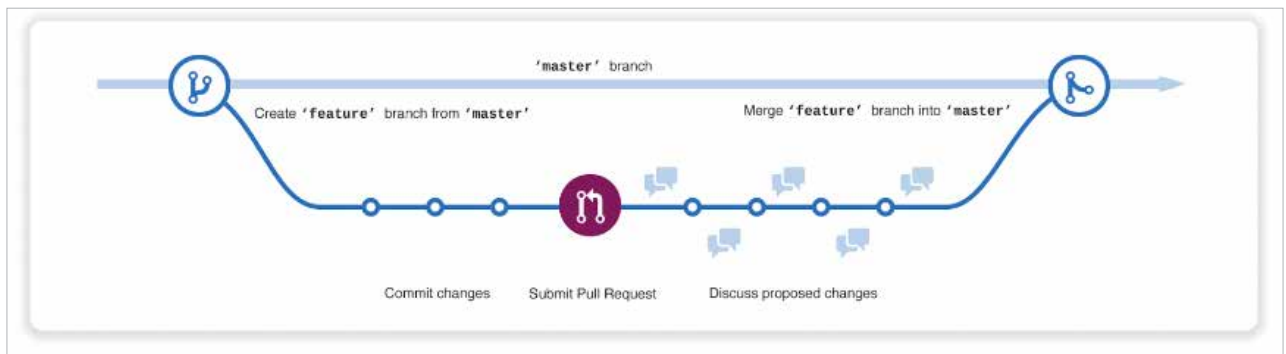


Figure 4: Pull request (PR in Git)

On GitHub, after pushing your `contact-form` branch to the remote repository, you create a pull request to merge it into the `main` branch that allows team members to review your code and suggest improvements before merging.

Clone: Cloning creates a copy of a remote repository on your local machine, enabling you to work on the project offline.

To work on an existing project hosted on GitHub, use `git clone https://github.com/user/repository.git` to create a local copy of the repository on your computer.

Push: Pushing sends your local commits to a remote repository. It's how you share your changes with others and

update the remote repo with your local work.

After committing changes locally, use `git push origin main` to upload your changes to the remote repository on GitHub. This updates the remote `main` branch with your local changes.

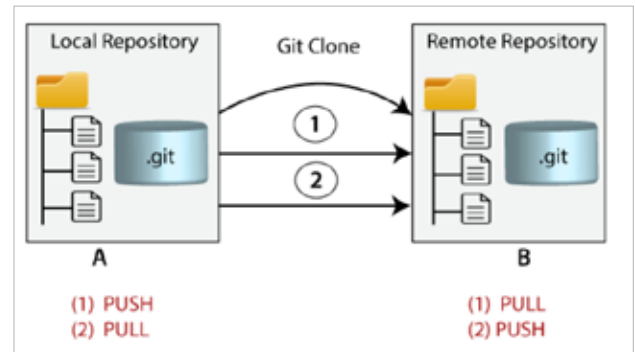


Figure 5: Clone (Git)

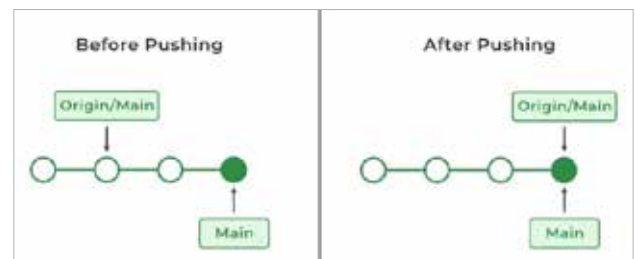


Figure 6: Pushing (Git)

Checkout: Checking out a branch or commit changes your working directory to reflect the state of that branch or commit.

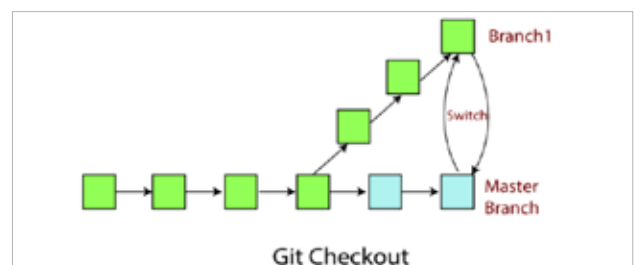


Figure 7: Checkout (Git)